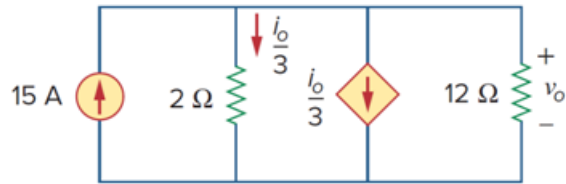


Problem

Find v_o and i_o in the circuit of Fig. 2.26.

Figure 2.26:



Step-by-step solution

Step 1 of 3

Refer to Figure 2.26 in the textbook.

Draw the circuit diagram with node notations.

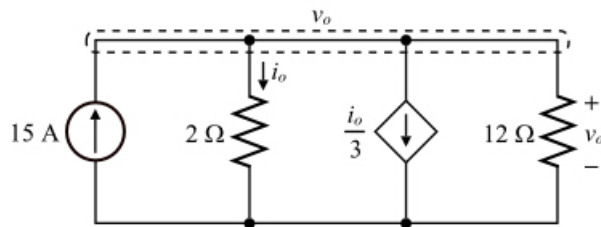


Figure 1

[Comments \(1\)](#)



Anonymous

Why does $i_o/3$ suddenly become i_o ?

Step 2 of 3

From the circuit shown in Figure 1, the current i_o is,

$$i_o = \frac{v_o}{2 \Omega} \dots\dots (1)$$

Apply Kirchhoff's current law at top node.

$$\frac{v_o}{2 \Omega} + \frac{i_o}{3} + \frac{v_o}{12 \Omega} = 15 \text{ A}$$

Substitute $\frac{v_o}{2}$ for i_o .

$$\frac{v_o}{2 \Omega} + \frac{v_o}{6 \Omega} + \frac{v_o}{12 \Omega} = 15 \text{ A}$$

$$\frac{6v_o + 2v_o + v_o}{12 \Omega} = 15 \text{ A}$$

$$9v_o = (15 \text{ A})(12 \Omega)$$

$$v_o = \frac{180 \text{ V}}{9}$$
$$= 20 \text{ V}$$

Therefore, the output voltage, v_o is **20 V**.

[Comments \(4\)](#)

☐

Anonymous

why is $i_o = v_o/2$? shouldnt it be $i_o = 3v_o/2$?

☐

Anonymous

you changed $i_o/3$ to i_o

☐

Anonymous

i guess the ans wrote that by mistake, but it doesnt matter if the 2 ohm branch is i_o or $i_o/3$ because the question is asking for V, (think of nodal analysis method, only resistor and volatge matters). you can try $1/3$, ans of V will be same.

☐

Anonymous

oh no nvm i was wrong. i dont get that either (

Step 3 of 3

Calculate the current i_o .

$$i_o = \frac{v_o}{2 \Omega}$$

Substitute 20 V for v_o .

$$i_o = \frac{20 \text{ V}}{2 \Omega}$$
$$= 10 \text{ A}$$

Therefore, the output current, i_o is **10 A**.